

इंटरनेट

मानक

Disclosure to Promote the Right To Information

Whereas the Parliament of India has set out to provide a practical regime of right to information for citizens to secure access to information under the control of public authorities, in order to promote transparency and accountability in the working of every public authority, and whereas the attached publication of the Bureau of Indian Standards is of particular interest to the public, particularly disadvantaged communities and those engaged in the pursuit of education and knowledge, the attached public safety standard is made available to promote the timely dissemination of this information in an accurate manner to the public.

“जानने का अधिकार, जीने का अधिकार”

Mazdoor Kisan Shakti Sangathan

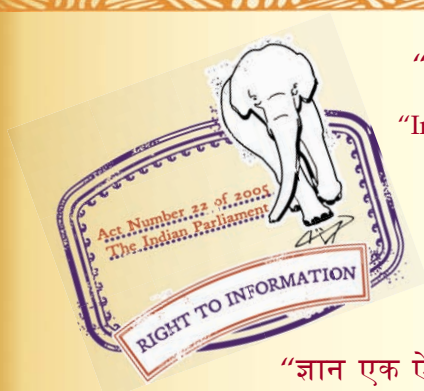
“The Right to Information, The Right to Live”

“पुराने को छोड़ नये के तरफ”

Jawaharlal Nehru

“Step Out From the Old to the New”

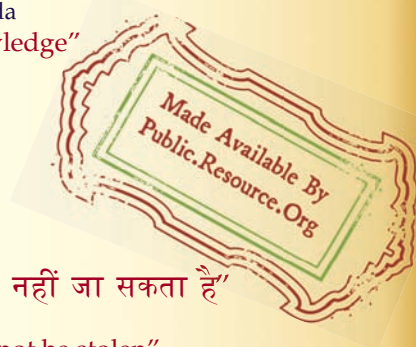
IS 13230 (1992): Graphical symbols for use on mine plans
[MED 8: Mining Techniques and Equipment]



“ज्ञान से एक नये भारत का निर्माण”

Satyanarayan Gangaram Pitroda

“Invent a New India Using Knowledge”



“ज्ञान एक ऐसा खजाना है जो कभी चुराया नहीं जा सकता है”

Bhartrhari—Nitiśatakam

“Knowledge is such a treasure which cannot be stolen”

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भारतीय मानक

खान योजनाओं में उपयोग के लिए ग्राफीय प्रतीक

Indian Standard

**GRAPHICAL SYMBOLS FOR USE ON
MINE PLANS**

UDC 622.1 : 003.62

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BHADUR SHAH ZAFAR MARG
NEW DELHI 110002

August 1992

Price Group 9

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Mine Sectional Committee had been approved by the Heavy Mechanical Engineering Division Council.

This standard is prepared to harmonising the graphical symbols for use on mine plans in mining industry.

Symbols and graphical representations are used to convey instructions and messages at a glance and considered to be quick methods of communication. Any ambiguity or misinterpretation of a symbol may lead to problem. Therefore symbols should be devised with utmost care to convey what exactly is to be conveyed.

The symbols may have to be devised for use at national level or company level or for specific area/field of use.

Indian Standard

GRAPHICAL SYMBOLS FOR USE ON MINE PLANS

1 SCOPE

This standard lays down the graphical symbols for use on mine plans required to be maintained by mine management under statutory requirements.

2 REFERENCES

IS No.	Title
696 : 1972	Code of practice for general engineering drawings
788 : 1971	Ink, drawing, waterproof, coloured
789 : 1971	Ink, drawing, waterproof, black
1064 : 1961	Paper sizes

3 RULES OF PRESENTATION

3.1 Classification

For ease of utilisation, symbols have been classified into two groups:

- Symbols for coal mines; and
- Symbols for metalliferous mines.

3.1.1 Individual symbols are specified in 6.

3.2 Size of Symbols

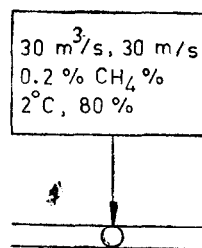
Size of symbols shall vary proportionately according to the scale used for the plan. In this standard, the size has been specified for the scale of 1 : 2 000.

3.3 Size of Letters

Letters used in the preparation of the mine plan shall be of 20, 14, 10, 7, 5, 3, and 2 mm size. Letters shall be written in the capital letters (except for standard abbreviation for units) for clear and easy comprehension on drawings.

3.4 When several measurements pertaining to a measurement point in the working are to be written on the mine plan, the point shall be indicated by a circle and quantities shall be enclosed in a rectangle.

For Example:



quantity and velocity of air methane

Temperature and relative humidity

3.5 Drawing Inks

The drawing inks used in the preparation of mine plans shall conform to IS 788 : 1971 and IS 789 : 1971. The colours specified in this standard are the same as specified in IS 788 : 1971.

4 SIZE OF MINE PLANS

The mine plans shall be prepared on drawing sheets of sizes specified for A-series of IS 1064 : 1961. For other aspects of drawing such as layout, numbering and referencing of sheets etc, the recommendations made by IS 696 : 1972 shall be followed as far as practicable.

5 SCALES

For preparation of various mine plans, the use of following scales is recommended:

1 : 200, 1 : 500, 1 : 1 000, 1 : 2 000, 1 : 5 000,
1 : 10 000, 1 : 20 000, 1 : 50 000, 1 : 100 000.

6 SYMBOLS

Symbols for coal mines and metalliferous mines are given in column 3 Table 1 and Table 2 respectively. Column 4 of Table 1 and Table 2 indicate the colour schemes and other additional information, if any, of these symbols.

Table 1 Symbols for Coal Mines
(Clause 6)

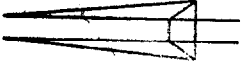
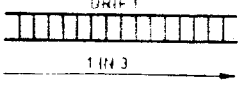
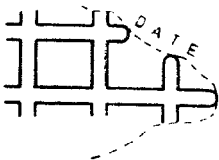
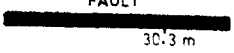
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
1	BOUNDARY OF LEASE HOLD		Red
120	UNDERGROUND COAL BARRIER		Green
3	SHAFT		—
4	ABANDONED SHAFT		—
5	INCLINE		—
6	ABANDONED INCLINE		—
7	PILLARS AND GALLERIES		—
8	DRIFT		Burnt sienna showing gradient in black
9	QUARTERLY SURVEY LINE		—
10	STAPLE SHAFT		Red, should state the distance up and down to all insets
11	ABANDONED STAPLE SHAFT		Red
12	FAULT		Red, showing the amount and direction of throw

Table 1 (continued)

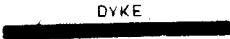
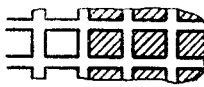
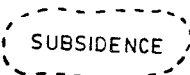
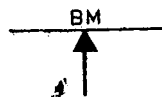
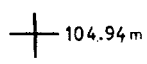
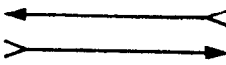
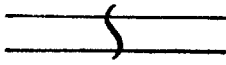
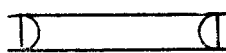
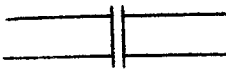
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
13	DYKE OR OTHER INTRUSION		Green
14	GOAF		—
15	SUBSIDENCE		Red
16	BENCH MARK		—
17	SURFACE CONTOUR		Burnt sienna
18	UNDERGROUND SPOT LEVEL		Blue
19	WATER DAM		Red
20	DIRECTION OF AIR CURRENT		Intake in blue Return in red
21	BRATTICE		Red
22	DOORS		Red
23	BRICKSTONE OR CONCRETE VENTILATION STOPPING		Red
24	FIRE DAM, SEAL OR STOPPING		Red
25	EXPLOSION PROOF STOPPING		Red

Table 1 (continued)

Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
26	AIR CROSSING		Black
27	EXPLOSION PROOF AIR CROSSING		Black
28	REGULATOR		Red
29	AUXILIARY FAN		Red
30	TELEPHONE		Red
31	UNDERGROUND FIRST AID STATION		Thick cross red
32	ENGINE HOUSE OR ROOM		—
33	BORE HOLE		Should show serial number and diameter
34	SURVEY LINES AND STATION		Red
35	SECTION OF SEAM		—
36	TEMPERATURE	—	Black
37	RELATIVE HUMIDITY	—	Black
38	AIR VELOCITY	—	Black

Table 1 (continued)

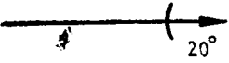
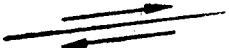
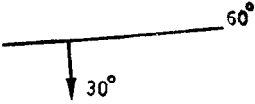
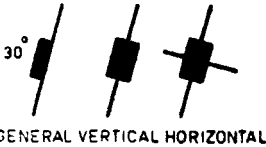
Symbol No.	Name	Symbol	Remarks
(1)	(2)	(3)	(4)
39	AIR FLOW	—	Black
40	GAS CONTENT	—	Scarlet
41	CONFIRMED GEOLOGICAL BOUNDARY	— — — — —	Black
42	INFERRED GEOLOGICAL BOUNDARY	- - - - -	Black
43	ARTICLINE		Prusian blue
44	SYNCLINE		Prusian bue
45	HORIZONTAL		Light green
46	STRIKE AND DIP OF STRATA		Black
47	THRUST FORMATION		Black
48	SCHISTOCITY OR FOLIATION	 GENERAL VERTICAL HORIZONTAL	Black
49	JOINT	 GENERAL VERTICAL HORIZONTAL	Black
50	TRAVELLING ROAD		Black and yellow
51	LADDER WAY		Black

Table 1 (continued)

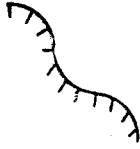
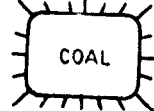
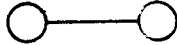
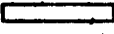
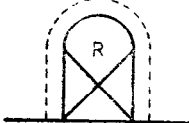
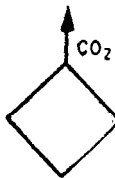
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
52	OPEN PIT		Black
53	STOCK PILE		Black
54	OLD WORKINGS		Black
55	PLANE TABLE TRANSVERSE LINES WITH STATIONS		Scarlet
56	THEODOLITE TRANSVERSE LINES WITH STATIONS		Black
57	TRIANGULATION STATIONS		Scarlet
58	LIMIT OF RESTRICTED AREA		Light green verged yellow
59	SAFETY PILLAR		Black line with scarlet shade
60	RESERVE STATION FOR BLASTING MATERIAL		Black
61	GAS EMISSION AREA/BLOWERS		Scarlet

Table 1 (continued)

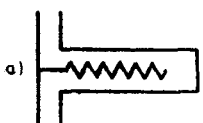
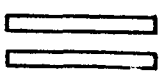
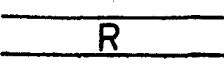
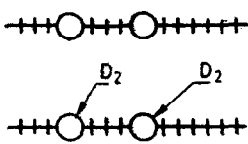
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
62	VENTILATION BRATTICE PARTITION	a) 	Scarlet
63	STONE DUST SPRAYED SITE		Yellow
64	STONE DUST BARRIER		Yellow
65	WATER SPRAYED AREA		Cobalt blue
66	FENCING		Scarlet
67	REGULATOR		Scarlet
68	LOADING STATION		Black
69	SHELTER		Light green
70	MINE TRACKS		Black
71	AERIAL ROPEWAY	a) 	Black
	b) MONOCABLE		—
72	DRILL		Black

Table 1 (continued)

Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
73	COAL CUTTER	 	Black
74	LOADER	 	Black
75	CONVEYOR	 	Black
76	HOIST/WINDER	  	Black
77	PIPE LINES		Light green
78	AIR COMPRESSOR	 	Black
79	BATTERY CHARGING STATION		Black
80	BOILER		Scarlet
81	FIRE EXTINGUISHER		Scarlet
82	WATER SUPPLY POINT		Black
83	TIPPLER		Black
84	TRAVERSER		Black

Table 1 (continued)

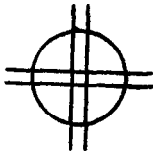
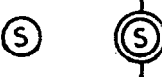
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
85	TURN TABLE		Black
86	POWER PLANT		Black
87	GENERATOR		Black
88	SUB-STATION		Black
89	MAIN SWITCH		Black
90	TRANSFORMER		Black
91	SWITCH		Black
92	ORDINARY ELECTRIC LAMP		Scarlet
93	FLAME PROOF SAFETY LAMP		Scarlet
94	TURBO LAMP		Scarlet
95	MOTOR		Black
96	HIGH TENSION CABLE		Scarlet
97	LOW TENSION CABLE OR CAB TYPE CABLE		Scarlet

Table 1 (continued)

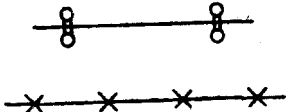
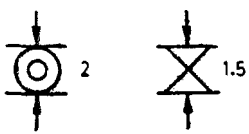
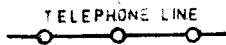
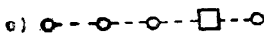
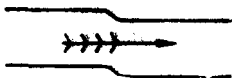
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
98	HIGH VOLTAGE LINE		Scarlet
99	SIGNAL LINE		Black
100	TELEPHONE CABLE		Cobalt blue
101	TELEGRAPH LINE		Black
102	TELEPHONE LINE		Black
103	ELECTRIC POWER LINE		Black
104	MINE BUILDING		Scarlet
105	RESCUE STATION		Cobalt blue
106	OIL STORAGE		Black with scar- lets for letters
107	WATER TANK		Cobalt blue
108	DRAIN		Cobalt blue
109	WATER RESERVOIR		Black line with cobalt blue shade
110	STREAM OR RIVER		Black line with cobalt blue shade arrow in black

Table 1 (continued)

Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
111	DRY NALA		Black line with cobalt blue shade
112	CANAL WITH SLUICE		Scarlet
113	WELL		Scarlet
114	WATER CLOSET		Black
115	BRIDGE		Black
116	RAILWAY LINE		Black
117	ROADS		Black
118	EMBANKMENT		Black
119	CUTTING		Black
120	SAND STORED AREA		Black
121	SPIRAL CHUTE		Black

Table 1 (concluded)

Symbol No.	Name	Symbol	Remarks
(1)	(2)	(3)	(4)
122	PROHIBITED AREA		—

Table 2 Symbols for Metalliferous Mines
(Clause 6)

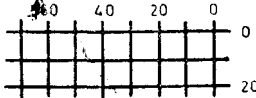
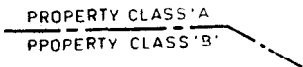
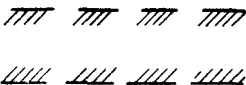
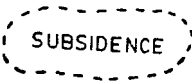
Symbol No.	Name	Symbol	Remarks
(1)	(2)	(3)	(4)
1	CO-ORDINATE LINES		Black
2	BOUNDARY OF MINING LEASEHOLD		Red
3	OUTCROP OF REET, VEIN, LODGE ETC.		Yellow
4	OUTCROP OF DYKE		Brown
5	SUBSIDENCE OF SURFACE		Red
6	PROHIBITED MINING AREA		Red
7	a) BOREHOLE (VERTICAL) FROM SURFACE		Red
	b) BOREHOLE (INCLINED) FROM SURFACE		Red

Table 2 (continued)

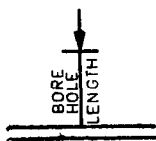
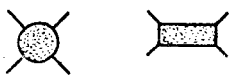
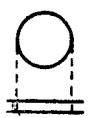
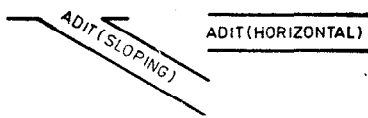
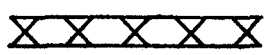
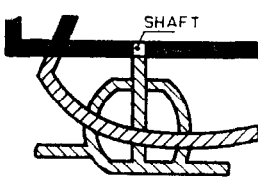
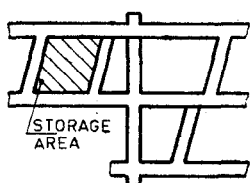
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
	c) BOREHOLES (UNDERGROUND)		Red
8	a) SHAFTS		Black
	b) ABANDONED SHAFTS		Black
9	WINZES		According to shape
10	ADIT		—
11	CROSS CUTS		—
12	LEVELS		—
13	VERTICAL PROJECTION OF UNDERGROUND WORKINGS		Black
14	a) PACK AND SAND FILLING		Black
	b) WASTE ROCK FILLING		Black

Table 2 (continued)

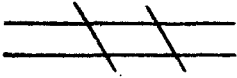
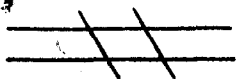
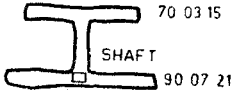
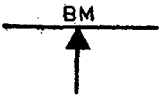
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
	c) GRANITE PACK WALLS		Black
15	a) FAULT PLANES		Green
	b) PEGMATITES INTRUSIONS		Blue
16	DYKE		Brown
17	DATE OF LAST SURVEY		Black
18	a) SURVEY STATION		Black
	b) PERMANENT STATION		Black
19	BENCH MARK		Black
20	a) BRICK, STONE OR CONCRETE VENTILATION STOPPINGS		Black
	b) HORIZONTAL SEALS		Black
21	BRATTICE		Red

Table 2 (continued)

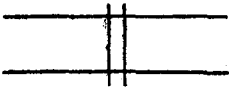
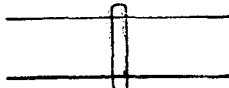
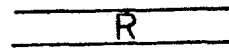
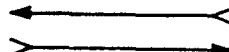
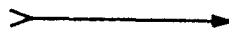
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
22	a) FIRE DAMS OR SEALS, OPEN		Red
	b) FIRE DAMS OR SEALS CLOSED		Red
23	WATER DAMS		Blue
24	a) DOORS OPEN		Black
	b) DOOR CLOSED		Black
25	REGULATORS		Red
26	AIR CROSSINGS		—
27	a) DIRECTION OF AIR CURRENT INTAKE		Blue
	b) RETURN		Red
28	AUXILIARY FAN		Black
29	TELEVISIONS		Green
30	UNDERGROUND AMBULANCE STATION		Red
31	FIRE REFUGE CHAMBER		Black

Table 2 (continued)

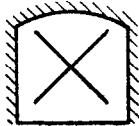
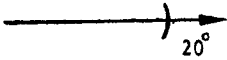
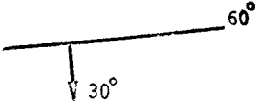
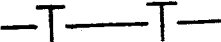
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
32	UNDERGROUND MAGAZINE		Black
33	PUMP STATION		Blue
34	TEMPERATURE	—	Black
35	RELATIVE HUMIDITY	—	Black
36	AIR VELOCITY	—	Black
37	AIR FLOW	—	Black
38	GAS CONTENT	—	Scarlet
39	CONFIRMED GEOLOGICAL BOUNDARY		Black
40	INFERRED GEOLOGICAL BOUNDARY		Black
41	ARTICLINE		Prusian blue
42	SYNCLINE		Prusian blue
43	HORIZONTAL		Light green
44	STRIKE AND DIP OF STRATA		Black
45	THRUST FORMATION		Black
46	SCHISTOCITY OR FOLIATION	 GENERAL VERTICAL HORIZONTAL	Black

Table 2 (continued)

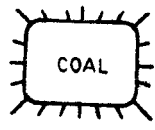
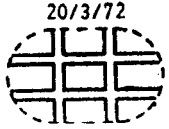
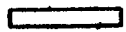
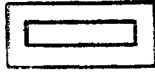
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
47	JOINT	 GENERAL VERTICAL HORIZONTAL	Black
48	TRAVELLING ROAD		Black and yellow
49	LADDER WAY		Black
50	OPEN PIT		Black
51	STOCK PILE		Black
52	OLD WORKINGS		Black
53	PLANE TABLE TRANSVERSE LINES WITH STATIONS		Scarlet
54	THEODOLITE TRANSVERSE LINES WITH STATIONS		Black
55	TRIANGULATION STATIONS		Scarlet
56	LIMIT OF RESTRICTED AREA		Light green verged yellow
57	SAFETY PILLAR		Black line with scarlet shade

Table 2 (continued)

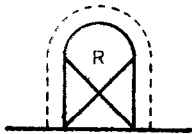
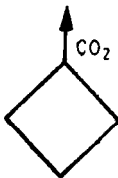
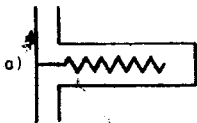
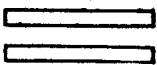
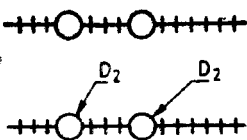
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
58	RESERVE STATION FOR BLASTING MATERIAL		Black
59	GAS EMISSION AREA/BLOWERS		Scarlet
60	VENTILATION BRATTICE PARTITION		Scarlet
61	STONE DUST SPRAYED SITE		Yellow
62	STONE DUST BARRIER		Yellow
63	WATER SPRAYED AREA		Cobalt blue
64	FENCING		Scarlet
65	REGULATOR		Scarlet
66	LOADING STATION		Black
67	SHELTER		Light green
68	MINE TRACKS		Black

Table 2 (continued)

Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
69	AERIAL ROPEWAY		
	a) BICABLE		Black
	b) MONOCABLE		—
70	DRILL	 	Black
71	COAL CUTTER	 	Black
72	LOADER	 	Black
73	CONVEYOR	 	Black
74	HOIST/WINDER	  	Black
75	PIPE LINES		Light green
76	AIR COMPRESSOR	 	Black
77	BATTERY CHARGING STATION		Black
78	BOILER		Scarlet
79	FIRE EXTINGUISHER		Scarlet
80	WATER SUPPLY POINT		Black

Table 2 (continued)

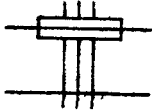
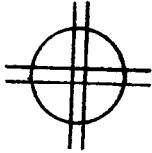
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
81	TIPPLER		Black
82	TRAVERSER		Black
83	TURN TABLE		Black
84	POWER PLANT		Black
85	GENERATOR		Black
86	SUB-STATION		Black
87	MAIN SWITCH		Black
88	TRANSFORMER		Black
89	SWITCH		Black
90	ORDINARY ELECTRIC LAMP		Scarlet
91	FLAME PROOF SAFETY LAMP		Scarlet

Table 2 (continued)

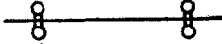
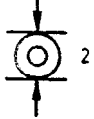
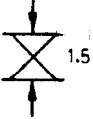
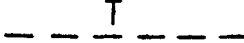
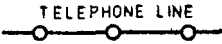
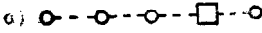
Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
92	TURBO LAMP		Scarlet
93	MOTOR	 	Black
94	HIGH TENSION CABLE		Scarlet
95	LOW TENSION CABLE OR CAB TYPE CABLE		Scarlet
96	HIGH VOLTAGE LINE	 	Scarlet
97	SIGNAL LINE	 	Black
98	TELEPHONE CABLE		Cobalt blue
99	TELEGRAPH LINE		Black
100	TELEPHONE LINE		Black
101	ELECTRIC POWER LINE		Black
102	MINE BUILDING		Scarlet
103	RESCUE STATION		Cobalt blue
104	OIL STORAGE		Black with scar- lets for letters

Table 2 (continued)

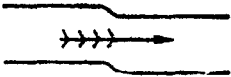
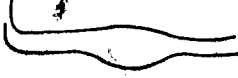
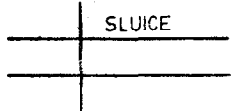
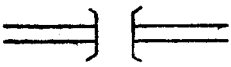
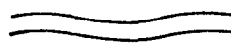
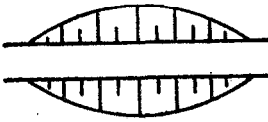
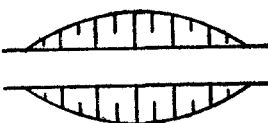
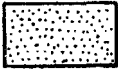
Symbol No.	Name	Symbol	Remarks
(1)	(2)	(3)	(4)
105	WATER TANK		Cobalt blue
106	DRAIN		Cobalt blue
107	WATER RESERVOIR		Black line with cobalt blue shade
108	STREAM OR RIVER		Black line with cobalt blue shade arrow in black
109	DRY NALA		Black line with cobalt blue shade
110	CANAL WITH SLUICE		Scarlet
111	WELL		Scarlet
112	WATER CLOSET		Black
113	BRIDGE		Black
114	RAILWAY LINE		Black
115	ROADS		Black
116	EMBANKMENT		Black
117	CUTTING		Black

Table 2 (concluded)

Symbol No. (1)	Name (2)	Symbol (3)	Remarks (4)
118	SAND STORED AREA		Black
119	SPIRAL CHUTE		Black
120	UNDERGROUND COAL		Green
121	STAPLE SHAFT		Red

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